

Suggested Software Improvement to Aid in Installation

The magnetometer is placed in the ground 100ft or more from the Raspberry Pi (located inside the station). For the person in the field installing the magnetometer, there is no convenient way to know if the magnetometer is oriented correctly.

As a suggested improvement to the runMag software, have the internet-connected Raspberry Pi report the XYZ data values to a cloud-based server. Since the Y-data represents the E vector, use the cloud-based server to convert the Y numerical data into an audible tone. Using a cellphone, or other device capable of reaching the internet, connect to a web page and have the tone relayed to the installer's personal device. While rotating the magnetometer and listening to the tone, the installer would be able to hear the tone pitch change as the unit is rotated to produce the minimum Y value response. At the null condition, the audio tone would be at its lowest pitch, and the E vector would be oriented toward the East. The H vector then, by default, would be oriented to produce the maximum Z value. In this scenario, the installer would only need to ensure that the unit remains level (as indicated by the bubble level) during rotation.